Credit Risk ML — OoS Model Validation Report

ILAB OeNB Project

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Executive Summary

This report validates 15 out-of-sample (OoS) credit risk models predicting corporate defaults in Austria. The models span 5 feature sets and 3 algorithms (XGBoost, AutoGluon, GLM), evaluated on approximately 70,000 firm-year observations with a 0.83% default rate.

Key Findings:

- **Best discrimination:** 04a_XGBoost (AUC = 0.920), closely followed by 03a_XGBoost (0.919)
- **Best calibration:** GLM and AutoGluon models (positive BSS); XGBoost is severely miscalibrated
- **Feature importance:** Time dynamics features provide +9 AUC points over static ratios
- **Early warning:** Models detect deterioration 4-5 periods before default (scores rise from 10% to 70%)
- **VAE adds minimal value:** Group 04 ≈ Group 03 performance despite additional latent features

1. Model Discrimination (AUC-ROC)

The AUC-ROC metric measures how well models rank defaults higher than non-defaults, regardless of calibration.

Interpretation:

- **Top tier (AUC > 0.90):** Models with time dynamics features (03a, 04a) across all algorithms
- **Middle tier (AUC 0.83-0.85):** VAE-only (05a) and financial ratios (02a)
- **Lower tier (AUC 0.81-0.82):** Raw balance sheet models (01a)

The ranking is dominated by **feature set choice**, not algorithm. Within each feature set, XGBoost leads slightly, followed by AutoGluon, then GLM — but the differences are small (1-2 AUC points).

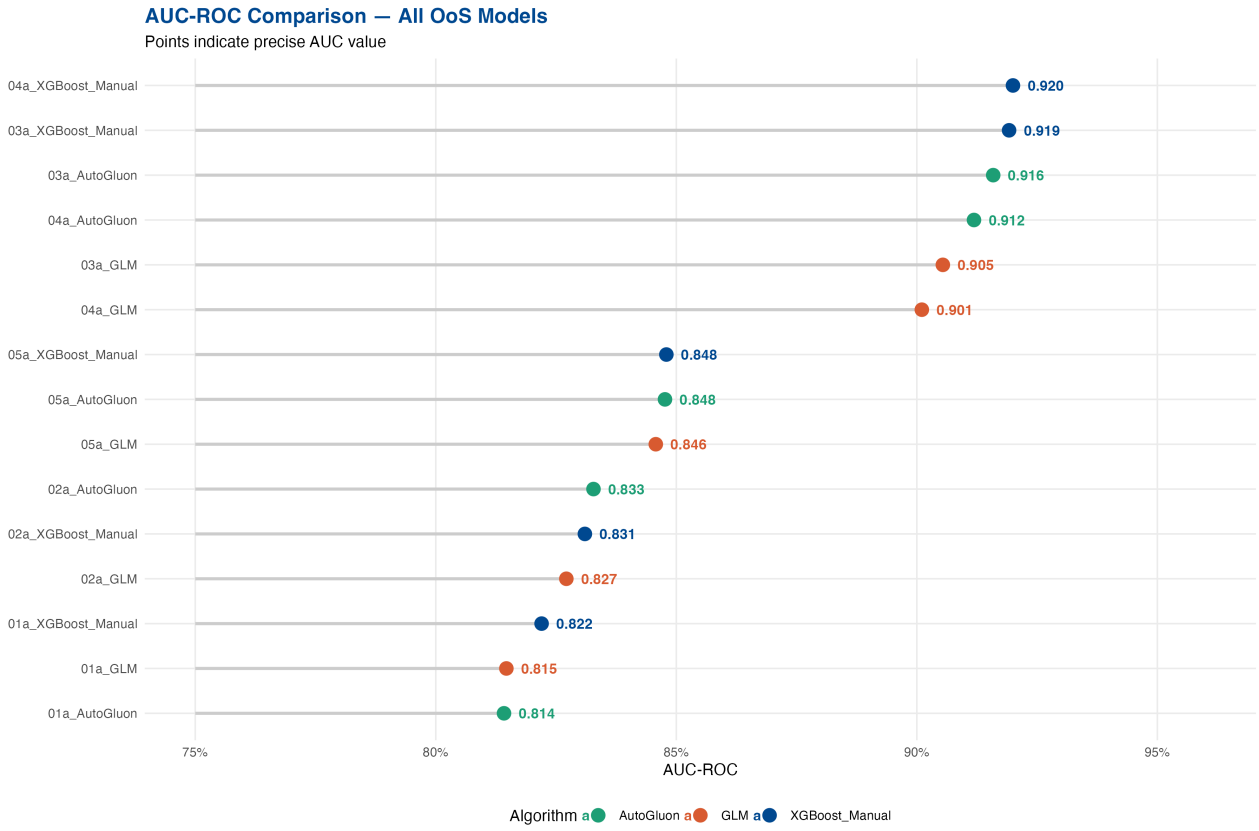

Figure 1: Model ranking by AUC-ROC. Time dynamics models (03a, 04a) dominate.

2. The Kolmogorov-Smirnov Statistic

The K-S statistic measures the maximum vertical distance between the cumulative distribution functions of default and non-default scores. Higher values indicate better separation between the two classes.

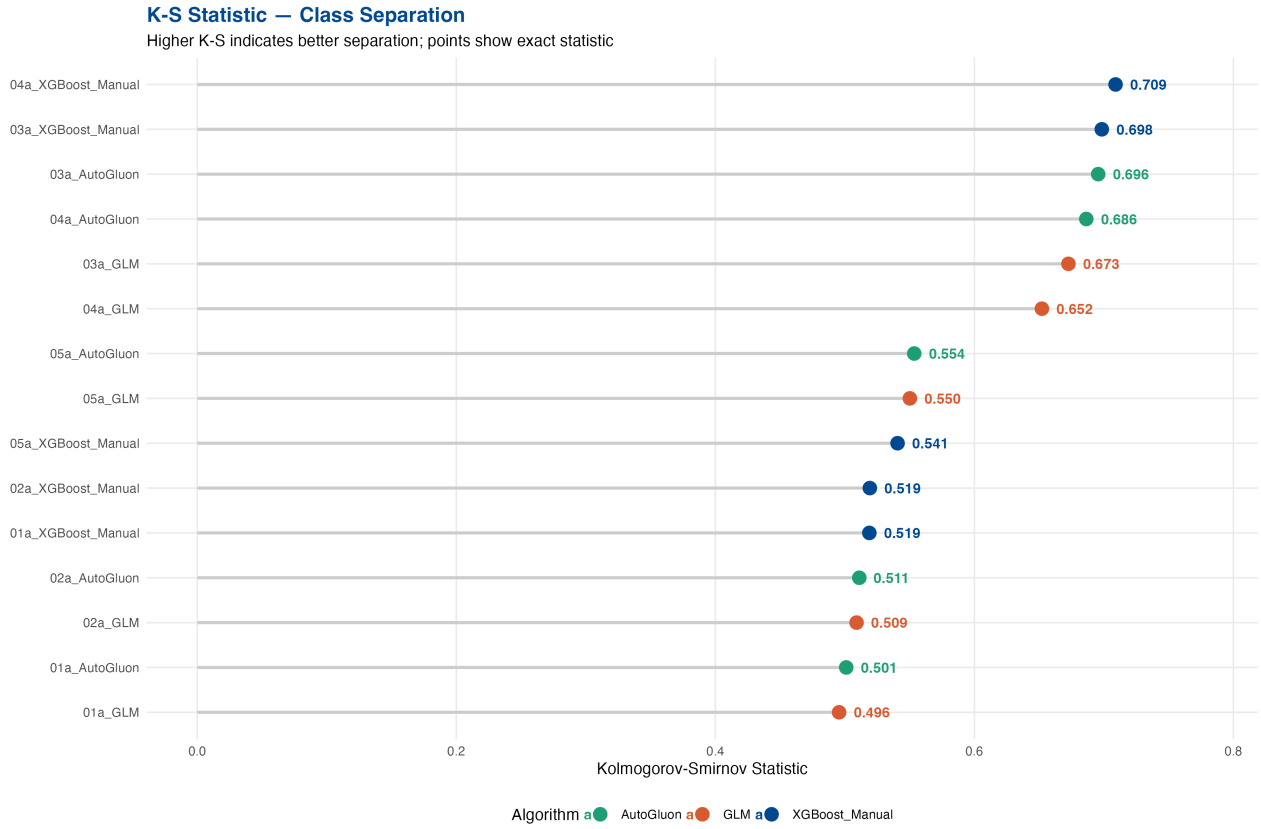

Figure 2: K-S statistic by model. Values above 0.5 indicate excellent separation.

What K-S tells us:

- **K-S > 0.65:** Excellent separation — the score distributions for defaults and non-defaults barely overlap
- **K-S 0.50-0.65:** Good separation — distinct distributions with some overlap
- **K-S < 0.50:** Moderate separation — substantial overlap between classes

Results:

- 03a_XGBoost achieves K-S = 0.698, meaning at the optimal threshold, we can separate 70% of defaults from non-defaults
- Time dynamics models (03a, 04a) consistently achieve K-S > 0.65
- Raw balance sheet models (01a) have K-S ≈ 0.50, showing the baseline is barely adequate

3. Calibration Analysis (Brier Skill Score)

Calibration measures whether predicted probabilities match observed frequencies. A model predicting 10% default probability should see 10% of those observations actually default.

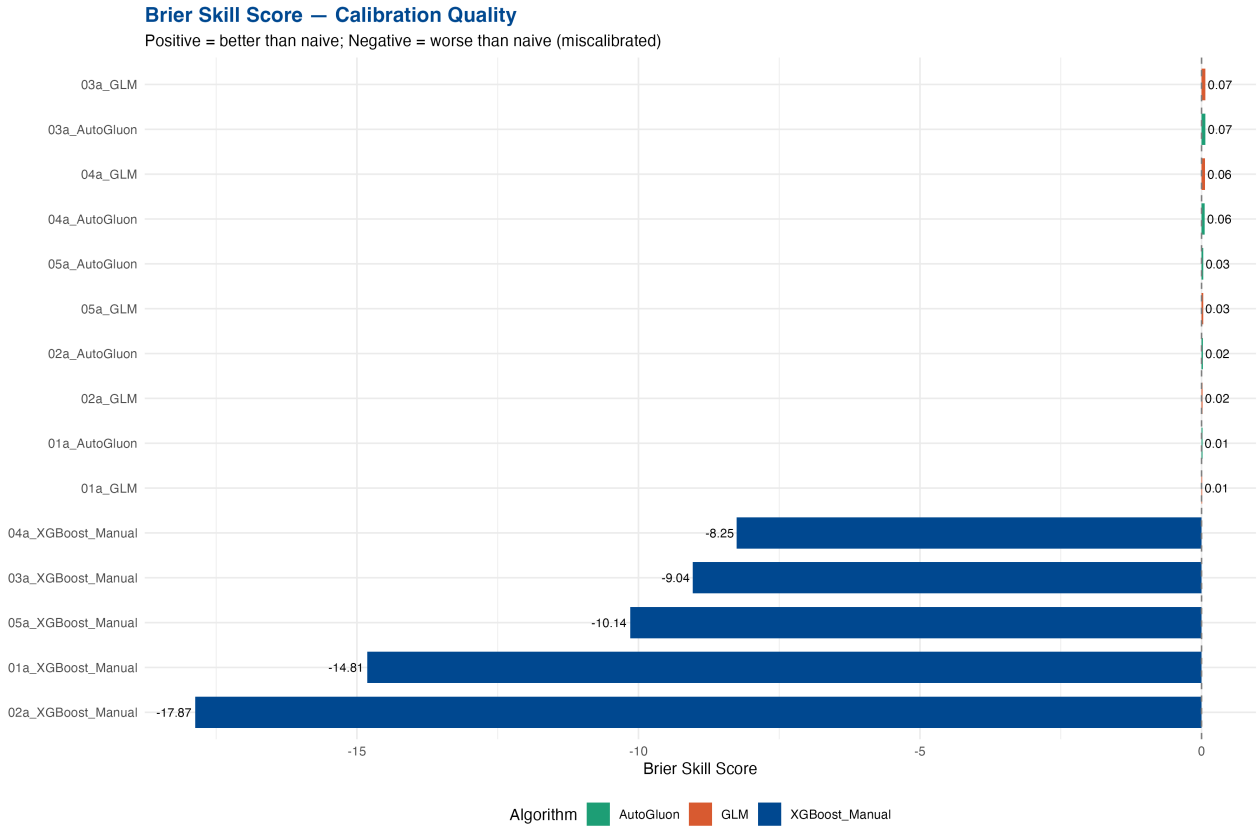

Figure 3: Brier Skill Score. Positive = better than naive baseline; Negative = worse (miscalibrated).

Critical finding: XGBoost is severely miscalibrated.

Algorithm	BSS Range	Interpretation
GLM	+0.01 to +0.07	Slightly better than naive — well-calibrated
AutoGluon	+0.01 to +0.07	Slightly better than naive — well-calibrated
XGBoost	**-8 to -18**	Much worse than naive — severely overconfident

Why XGBoost is miscalibrated:

XGBoost predicts probabilities of 60-70% for the highest-risk decile, but actual default rates are only 3-4%. The model is overconfident by a factor of 20×. This is a known issue with tree-based models that optimize for ranking rather than probability estimation.

Recommendation: If using XGBoost predictions for business decisions requiring accurate probabilities (pricing, capital allocation), apply Platt scaling or isotonic regression to recalibrate.

4. Calibration Curves by Feature Set

The calibration plot shows predicted probability (x-axis) vs. observed default rate (y-axis) by decile. Points on the diagonal indicate perfect calibration.

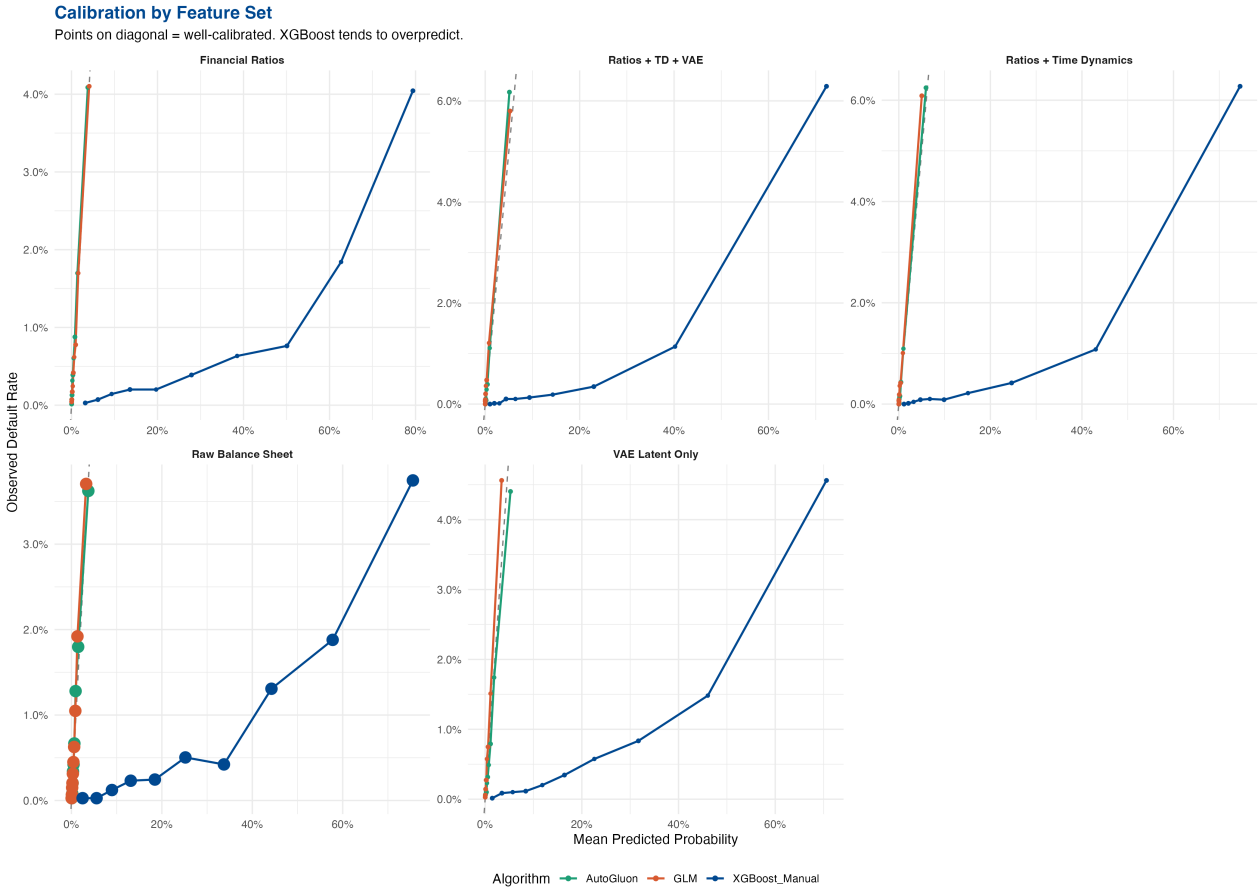

Figure 4: Calibration curves. GLM/AutoGluon hug the diagonal; XGBoost curves sharply upward.

Pattern explanation:

- **GLM and AutoGluon (green/coral):** Predictions stay in the 0-10% range, closely tracking observed rates
- **XGBoost (blue):** Predictions spread from 0-80%, but observed rates only reach 4-6%

This confirms XGBoost's discrimination-calibration trade-off: it creates wider score separation (better ranking) but sacrifices probability accuracy.

5. Score Separation Between Classes

This plot shows the mean predicted probability for defaults (coral) vs. non-defaults (blue) for each model. Larger gaps indicate better discrimination.

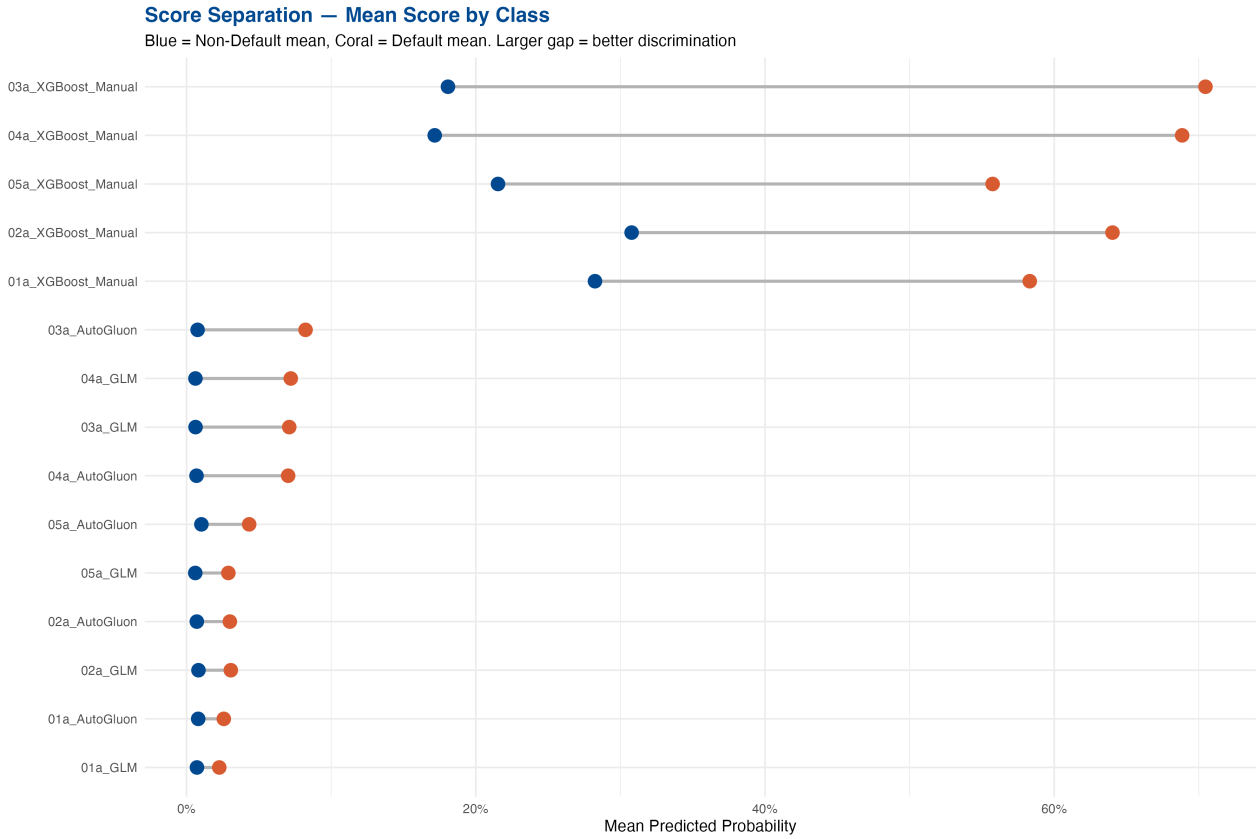

Figure 5: Mean scores by class. XGBoost creates the largest separation but is miscalibrated.

Key insights:

- **XGBoost models:** Non-defaults average 17-31%, defaults average 56-70% — a 40-50 point gap
- **GLM/AutoGluon:** Non-defaults average 0.6-1%, defaults average 2-8% — a 2-7 point gap

Both patterns achieve similar AUC because AUC measures *relative* ranking, not absolute score spread. GLM's compressed scores are actually better calibrated to the true 0.83% base rate.

6. Score Distributions by Model

The density plots show the full distribution of predicted probabilities for each class.

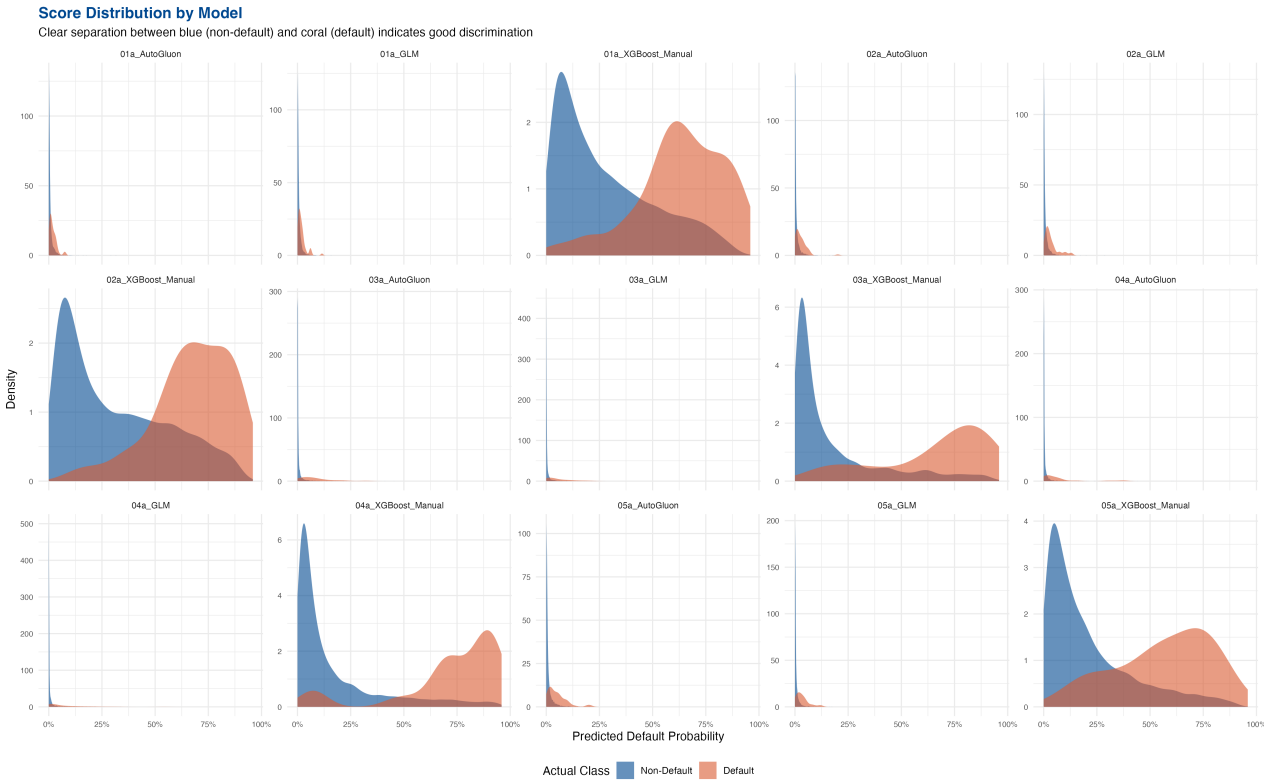

Figure 6: Score distributions. Clear separation between blue (non-default) and coral (default) indicates good discrimination.

Visual patterns:

- **XGBoost models:** Wide, overlapping distributions with distinct peaks
- **GLM/AutoGluon:** Narrow, spike-shaped distributions concentrated near zero
- **Best separation:** 03a and 04a models show the clearest distinction between classes

The bimodal shape in XGBoost's default distribution (coral) suggests the model identifies two subpopulations: firms showing gradual deterioration vs. sudden defaults.

7. ROC Curves by Feature Set

ROC curves plot the trade-off between true positive rate (sensitivity) and false positive rate.

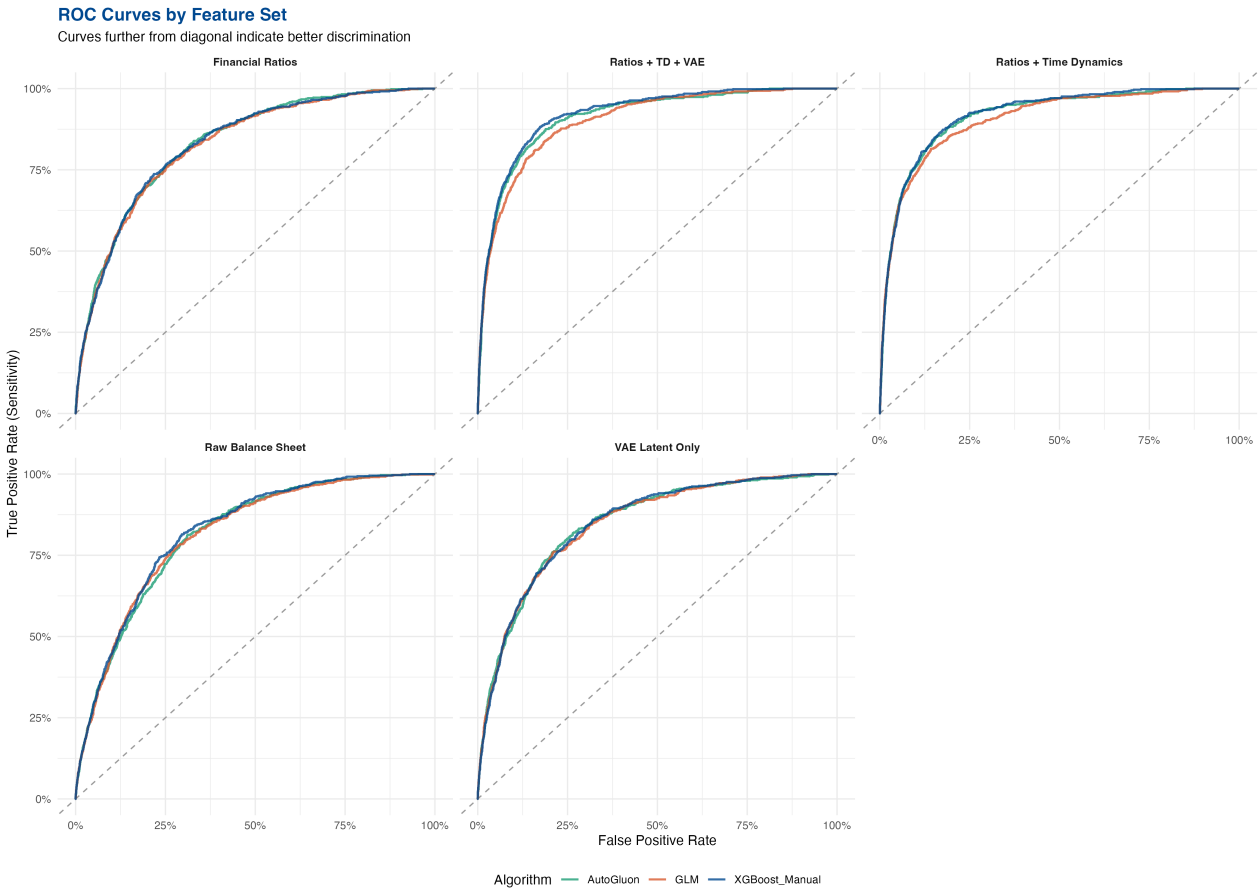

Figure 7: ROC curves. Ratios + Time Dynamics achieves the best trade-off.

Observations:

- **Ratios + Time Dynamics** and **Ratios + TD + VAE** show nearly identical, excellent ROC curves
- **Financial Ratios** and **Raw Balance Sheet** have more modest performance
- Within each feature set, all three algorithms perform similarly (curves nearly overlap)

This confirms that **feature engineering dominates algorithm choice** in credit risk modeling.

8. Feature Set Comparison

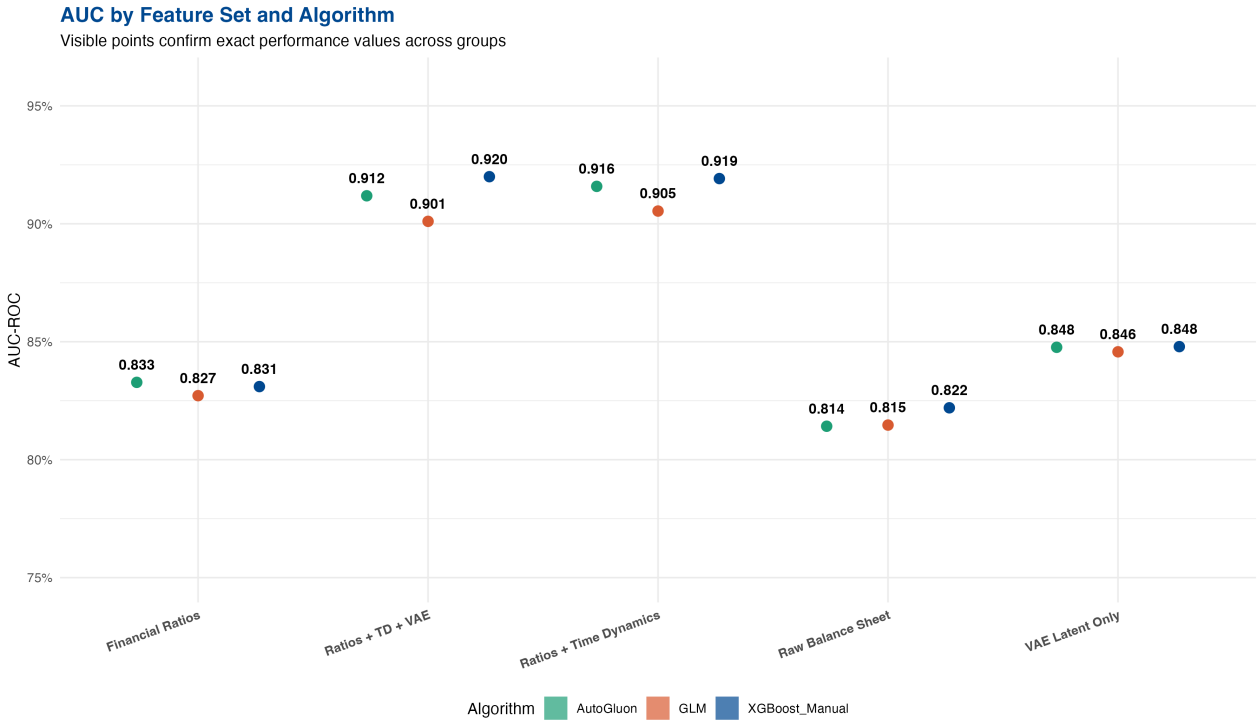

Figure 8: AUC by feature set and algorithm. Time dynamics provides the largest performance boost.

Feature set ranking:

Rank	Feature Set	Mean AUC	Gain vs. Raw
1	Ratios + TD + VAE	0.911	+9.6 pts
2	Ratios + Time Dynamics	0.914	+9.9 pts
3	VAE Latent Only	0.847	+3.2 pts
4	Financial Ratios	0.831	+1.6 pts
5	Raw Balance Sheet	0.817	baseline

Why time dynamics helps:

Time dynamics features capture *deterioration trajectories* — exponential moving averages, momentum, peak-drop metrics. A firm that has always been weak looks different from one actively declining. This temporal signal is the strongest predictor of default.

Why VAE adds little value:

The VAE latent features encode compressed financial structure, but this information is largely redundant with the sector-relative ratios (`secRank_*`) already in the data. Groups 03 and 04 perform nearly identically.

9. Early Warning Capability (Score Trajectory)

The trajectory analysis tracks how predicted probabilities evolve for firms that eventually default.

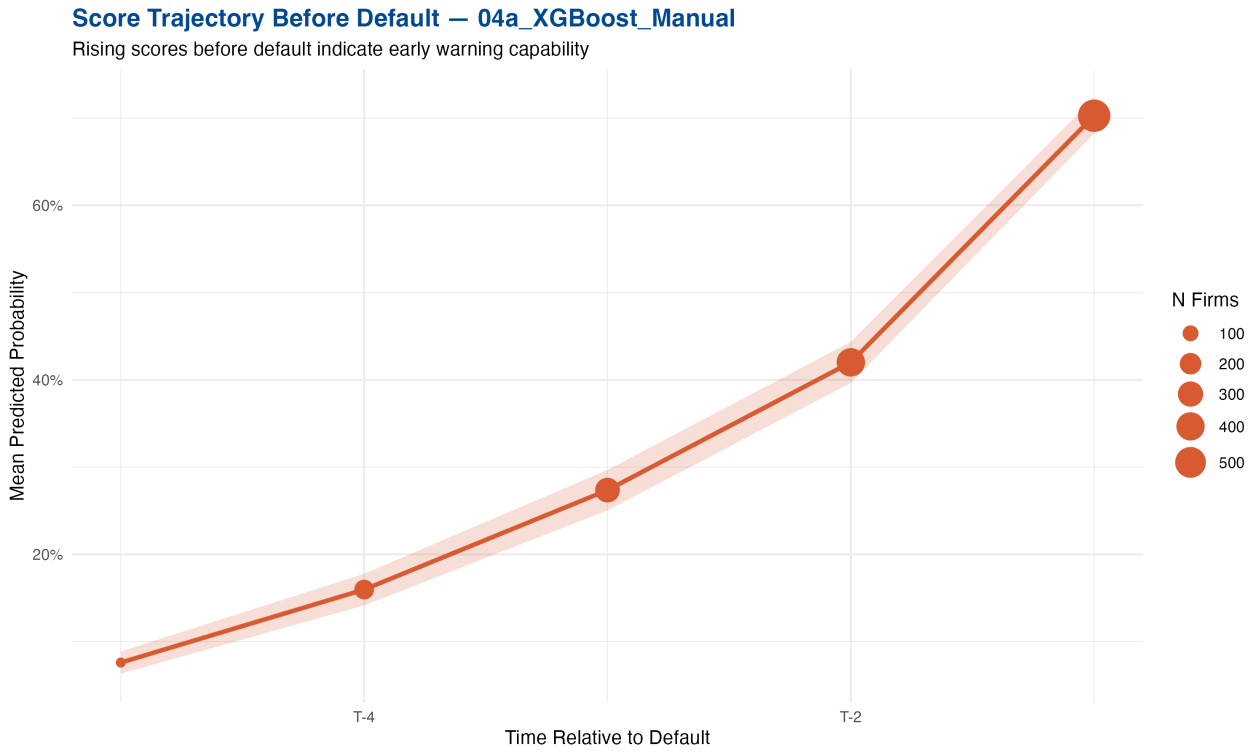

Figure 9: Score trajectory for defaulting firms. Rising scores indicate early warning capability.

This is a critical validation result.

The best model (04a_XGBoost) shows scores rising from ~10% at T-5 to ~70% at default — a 7× increase over 5 periods. This demonstrates genuine early warning capability:

- **T-5:** Firms already show elevated risk (10% vs. 17% population average)
- **T-3:** Clear deterioration visible (28%)
- **T-1:** High alert (42%)
- **Default:** Model is highly confident (70%)

Implication: The model doesn't just identify high-risk firms — it tracks their deterioration trajectory, enabling proactive intervention.

10. Why Some Models Perform Better

Feature Set is the Primary Driver

The validation conclusively shows that **feature engineering matters more than algorithm choice**:

- Adding time dynamics to financial ratios: **+8.3 AUC points**
- Switching from GLM to XGBoost within same features: **+1.4 AUC points**

The time dynamics features work because corporate defaults are processes, not events. Firms deteriorate over multiple reporting periods before failing. Features like `history_length`, `expmean_r*` (exponential moving averages), and `peak_drop_r*` capture this trajectory.

Algorithm Trade-offs

Algorithm	Strengths	Weaknesses
XGBoost	Best discrimination, highest AUC	Severely miscalibrated probabilities
AutoGluon	Good discrimination, well-calibrated	Slightly lower AUC than XGBoost
GLM	Best calibration, interpretable	Lowest discrimination